Part A — Theory: Classification in Machine Learning

An academic overview of classification concepts, algorithms, evaluation metrics, and a real-world application

1. Introduction to Classification

Classification is a category of supervised learning in which a model is trained to assign each input observation to one of a fixed set of predefined categories, or classes. During training, the algorithm is shown many labeled examples and learns a decision function that maps the input features to a discrete output label; once trained, it applies this same function to unseen data in order to predict the class to which each new observation most likely belongs.

The key distinction between classification and regression lies in the type of output variable each problem produces. **Classification** problems have a categorical target — the answer is one label chosen from a finite, discrete set (for example, "spam" or "not spam"). **Regression** problems, in contrast, have a continuous numerical target — the model outputs a value that can, in principle, take any number along a range, such as a price, a temperature, or a probability of rainfall in millimetres. Put differently, classification answers the question "which category?" while regression answers the question "how much, or how many?".

A concrete example of classification is a bank's fraud-detection system, which examines the details of a transaction and labels it as either "fraudulent" or "legitimate." A corresponding example of regression is a real estate pricing model, which takes a property's location, floor area, and number of bedrooms as input and outputs a continuous predicted sale price in dollars.

2. Classification Algorithms

Three widely used classification algorithms are logistic regression, decision trees, and random forests. Although all three are designed to solve the same underlying task, they differ substantially in how they arrive at a prediction, in how easy they are to interpret, and in the situations where they tend to perform best.

2.1 Logistic Regression

Logistic regression works by fitting a linear combination of the input features and then passing that combination through a sigmoid (logistic) function, which compresses the result into a probability between 0 and 1. If the resulting probability exceeds a chosen threshold — commonly 0.5 — the observation is assigned to the positive class; otherwise it is assigned to the negative class. Because the coefficient attached to each feature indicates both the direction and the strength of that feature's influence on the outcome, the model is easy to interpret and explain. A typical real-world application is credit scoring, where a lender estimates the likelihood that an applicant will default on a loan. The main strength of logistic regression is that it is fast to train, computationally inexpensive, and transparent, but its central limitation is that it assumes the classes can be separated by a roughly linear boundary, so it tends to underperform when the true relationship between the features and the outcome is highly non-linear.

2.2 Decision Trees

A decision tree builds a prediction model by repeatedly splitting the dataset into smaller and smaller subsets according to feature values, forming a tree-like structure of yes/no decision rules. At each split, the algorithm chooses the feature and threshold that most effectively separates the classes, typically by minimizing an impurity measure such as the Gini index or maximizing information gain based on entropy. Once training is complete, a new observation is classified by walking it down the tree, from the root to a leaf, following whichever branch matches its feature values at each step. A common real-world use case is clinical decision support, where a tree-based tool guides a physician through a sequence of symptom checks to arrive at a probable diagnosis. Decision trees are valued because they are intuitive, can be visualized directly, mirror the way humans naturally reason through a sequence of choices, and require no scaling of the input features. Their major weakness is a strong tendency to overfit: an unconstrained tree can grow deep enough to memorize noise in the training data rather than capturing the underlying pattern, which hurts its ability to generalize to new cases.

2.3 Random Forest

A random forest is an ensemble method that trains a large collection of individual decision trees and then combines their outputs, typically through a majority vote for classification tasks. Diversity among the trees is introduced through bagging (bootstrap aggregating), where each tree is trained on a randomly sampled subset of the rows, and through random feature selection, where each split only considers a random subset of the available features. This combination means that no single tree dominates the final prediction, and the errors of individual trees tend to cancel out. A practical application is customer churn prediction in e-commerce, where a company wants to identify which subscribers are likely to cancel their service. The main advantage of a random forest over a single decision tree is a substantial reduction in overfitting together with strong accuracy on complex, high-dimensional data, but this comes at the cost of slower prediction speed, higher memory consumption, and reduced interpretability, since it is difficult to trace a single prediction back through hundreds of individual trees.

In short, logistic regression offers speed and interpretability at the cost of flexibility; decision trees offer interpretability and flexibility at the cost of stability; and random forests offer strong predictive accuracy and stability at the cost of interpretability and computational efficiency.

3. Classification Metrics

Once a classifier has been trained, its quality must be measured using metrics that summarize how well its predictions match the true labels.

Accuracy is the proportion of all predictions, across both classes, that were correct. It gives a single, easy-to-understand summary of overall performance, but it treats every class as equally important.

Precision measures, out of all the cases the model predicted as positive, how many were genuinely positive. It reflects how trustworthy a positive prediction from the model actually is, and it is most useful when the cost of a false positive is high.

Recall (sensitivity) measures, out of all the cases that were genuinely positive, how many the model successfully identified. It reflects the model's ability to avoid missing real positive cases, and it matters most when the cost of a false negative is high.

F1-Score is the harmonic mean of precision and recall. Because the harmonic mean penalizes extreme imbalance between the two, a high F1-score can only be achieved when precision and recall are both reasonably strong, making it a useful single figure when a balance between the two is desired.

Confusion Matrix is a simple two-by-two table (for binary classification) that lays out the four possible prediction outcomes — true positives, true negatives, false positives, and false negatives — allowing accuracy, precision, recall, and F1-score to all be derived directly from its cells.

Table 1. Comparison of classification evaluation metrics.

Metric	What It Measures	Sensitivity to Class Imbalance
Accuracy	The overall proportion of predictions the model got right.	Very high — a large majority class can make accuracy look excellent even when the minority class is barely detected.
Precision	Of everything the model labeled positive, how many actually were positive.	Low — it reflects only the cost of false positives, not how the classes are distributed.
Recall	Of everything that was actually positive, how many the model correctly caught.	Low — it reflects only the cost of false negatives, not class distribution.
F1-Score	The harmonic mean of precision and recall, balancing both concerns in one number.	Resilient — remains informative even when one class heavily outweighs the other.

4. Class Imbalance

Accuracy can be dangerously misleading whenever one class greatly outnumbers the other. For instance, if a loan dataset contains 95 percent approved applications and only 5 percent rejected ones, a trivial model that predicts "approved" for every single applicant, regardless of their actual risk, will still achieve a 95 percent accuracy score. Despite this seemingly impressive number, such a model has learned nothing useful: it fails to identify a single high-risk applicant and is therefore worthless for the bank's actual purpose of managing default risk.

Precision is prioritized when the cost of a false positive is severe. In a loan-approval context, a false positive means approving an applicant who will ultimately default, which can translate into a substantial direct financial loss for the lender. A bank that is primarily concerned with protecting its capital will therefore favour a model with high precision, even if that model ends up rejecting some applicants who would actually have repaid.

Recall is prioritized when the cost of a false negative is severe. If a lender's strategic goal is aggressive portfolio growth and it is more concerned with rejecting good customers than with occasionally approving a risky one, or if the same reasoning is applied outside finance — for instance, a medical screening test where missing a genuine case of a life-threatening disease is far worse than a false alarm — then recall becomes the metric to prioritize, since the primary goal is to avoid missing positive cases at almost any cost.

5. Real-World Case Study

A relevant example of classification applied in practice comes from the field of microfinance credit risk assessment. Financial institutions serving low-income or previously unbanked customers often lack the extensive credit history that traditional scorecards rely on, so several microfinance providers and fintech companies have turned to machine learning classifiers built on alternative data sources instead.

Goal: The objective of such a project is typically to predict, at an early stage, which borrowers are likely to default on a microloan, so that the lender can adjust loan terms or decline high-risk applications before capital is disbursed.

Data Used: These systems commonly draw on anonymized transaction histories, basic demographic indicators, requested loan amounts, and behavioural signals such as mobile phone airtime top-up patterns or past repayment records, gathered from a customer base numbering in the thousands.

Classifier Applied: A random forest classifier is a natural fit for this kind of problem, since the available features are a mix of numerical and categorical variables with complex, non-linear interactions, and the model's ensemble structure helps guard against overfitting to a relatively small and noisy dataset. Hyperparameters such as the number of trees and maximum tree depth are typically tuned through a grid or randomized search.

Key Results and Insights: In studies of this type, ensemble classifiers have been shown to outperform traditional, rule-based credit scorecards, often improving the area under the ROC curve from roughly the low-0.70s to the mid-0.80s. A recurring insight is that short-term, behavioural indicators of repayment consistency tend to carry more predictive power than static demographic attributes, suggesting that how a customer manages recent, small financial obligations is a stronger signal of future creditworthiness than who they are on paper. This kind of result illustrates the broader value of classification in finance: it allows an institution to move from blanket, one-size-fits-all lending decisions toward risk assessments that are tailored to the individual applicant.

References

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